

**INT 374: DATA ANALYTICS WITH POWER
BI PROJECT REPORT**

(Project Semester October 2025 -January 2026)



L OVELY
P ROFESSIONAL
U NIVERSITY

Credit Card Ecosystem Analysis

Submitted by

Sudeep Kumar Reddy Eaga

Registration No: 12318529

Section: KM003

Course Code: INT 374

Under the Guidance of

Dr. Madhu Bala

31770

Discipline of CSE/IT

Lovely School of Computer Science

Lovely Professional University, Phagwara

CERTIFICATE

This is to certify that Sudeep Kumar Reddy Eaga bearing Registration no. 12318529 has completed INT 374 project titled, “**Credit Card Ecosystem Analysis**” under my guidance and supervision. To the best of my knowledge, the present work is the result of his/her original development, effort and study.

Signature and Name of the Supervisor

Designation of the Supervisor

School of Computer Science.

Lovely Professional University

Phagwara, Punjab.

Date:

DECLARATION

I, Sudeep Kumar Reddy Eaga, student of B-tech Computer Science and Engineering under CSE/IT Discipline at, Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine.

Date: 14-11-2025

Signature: Sudeep Kumar Reddy Eaga

Registration No: 12318529

Table Of Contents

1. [Introduction](#)
2. [Source of Dataset](#)
3. [Preprocessing of Data](#)
4. [Model Building](#)
5. [Analysis on Dataset](#)
6. [Analysis of Dashboards](#)
 - 6.1 Objective 1: To implement a snowflake schema data model in Power BI
 - 6.2 Objective 2: To analyze customer spending behavior and financial patterns
 - 6.3 Objective 3: To identify top-performing merchants and geographic revenue trends
 - 6.4 Objective 4: To monitor fraud and risk patterns using customer and card indicators
 - 6.5 Objective 5: To evaluate card performance across different card types and brands
 - 6.6 Objective 6: To build an interactive Power BI dashboard for decision-making
7. [Conclusion](#)
8. [Future Scope](#)
9. [Reference](#)

1. Introduction

In today's digital banking environment, financial institutions generate massive volumes of transactional data that require effective analytics for business intelligence, customer insights, and fraud prevention. This project focuses on end-to-end credit card ecosystem analytics using Power BI, leveraging a real-world transactions and fraud dataset to deliver actionable insights through interactive dashboards.

The project integrates multiple data dimensions, including transactions, cards, users, merchants, income categories, geographic locations, and risk indicators, using a well-structured star-schema data model. Relationships between fact and dimension tables enable accurate aggregations and drill-down analysis across customer behaviour, merchant performance, and fraud risk patterns.

Key Analytical Objectives:

- Analyze customer spending behaviour across age groups, gender, income categories, and card types
- Identify top-performing merchants and states based on transaction volume and revenue contribution
- Evaluate card performance by card brand and card type using balance and credit limit metrics
- Monitor fraud and risk patterns using indicators such as high-risk flags, credit scores, and chip-enabled transactions
- Understand the relationship between credit score, card usage, and fraud risk

Dashboards Developed:

1. Customer Insight Dashboard

- Balance distribution by age group and gender
- Average debit amount by income category
- Geographic visualization of customer spending
- Demographic-driven financial behaviour analysis

2. Merchant & Spend Category Analytics Dashboard

- Top states with highest merchant revenue
- Top 15 merchants contributing maximum balance
- Merchant-wise and location-wise revenue insights

3. Card Performance Dashboard

- Credit limit analysis by card type and brand
- Balance distribution across debit, credit, and prepaid cards
- Performance comparison of Amex, Visa, Mastercard, and Discover

4. Fraud & Risk Monitoring Dashboard

- Transactions analysis based on chip-enabled cards
- Credit score comparison between normal and high-risk customers

- Card brands and card types flagged as high risk
- Risk-focused segmentation for proactive fraud monitoring

Tools & Techniques Used:

- Power BI for data modelling, DAX calculations, and interactive dashboards
- Power Query for data cleaning, transformation, and preprocessing
- DAX Measures for KPIs such as total balance, average credit score, transaction counts, and risk indicators
- Geospatial analysis using map visuals for location-based insights

Outcome:

This project demonstrates how data-driven analytics can enhance decision-making in the banking and financial services domain. By combining customer insights, merchant performance analysis, card usage trends, and fraud risk monitoring into a unified Power BI solution, the dashboards enable stakeholders to identify revenue opportunities, assess customer value, and proactively manage financial risk.

2. Source of dataset

The dataset used in this project is sourced from Kaggle, a widely recognized open platform for data science and analytics projects. The dataset is titled “Transactions Fraud Datasets” and is publicly available at:

 <https://www.kaggle.com/datasets/computingvictor/transactions-fraud-datasets>

This dataset contains comprehensive information related to credit card transactions, customer demographics, card details, merchant activity, and fraud risk indicators. It is designed to support analytical use cases such as customer behaviour analysis, merchant performance evaluation, card usage monitoring, and fraud risk assessment.

Key attributes in the dataset include:

- Transaction details: transaction ID, transaction date, transaction amount (Balance_\$), quarter
- Customer information: age, age group, gender, income category, credit score, number of credit cards
- Card attributes: card ID, card brand (Visa, Mastercard, Amex, Discover), card type (Credit, Debit, Debit Prepaid), credit limit, chip availability
- Merchant details: merchant ID, merchant state, merchant city, ZIP code, MCC (merchant category code)
- Risk indicators: high-risk flag, fraud-related signals.
- Location data: latitude and longitude for geographical analysis

The dataset is well-suited for business intelligence and descriptive analytics, enabling deep insights into:

- Customer spending behaviour across age groups and income categories
- Card performance by card type and brand
- Merchant revenue distribution across states and locations
- Fraud and risk monitoring based on card type, chip availability, and credit score patterns

In this project, the data is modelled using a star schema in Power BI, with a central transaction’s fact table connected to multiple dimension tables such as users, cards, merchants, and location data. This data model supports efficient filtering, slicing, and drill-down analysis across dashboards.

The dataset was downloaded in CSV format and used strictly for academic and educational purposes. Before visualization, extensive data preprocessing and transformation were performed using Power Query, including:

- Handling missing and inconsistent values
- Creating derived columns such as Age Group, Income Category, and High-Risk Flag
- Standardizing categorical fields
- Aggregating transactional metrics

- Establishing relationships between fact and dimension tables

These preprocessing steps ensured high data quality, improved analytical accuracy, and enabled the development of interactive Power BI dashboards such as:

- Customer Insight Dashboard
- Merchant & Spend Category Analytics Dashboard
- Card Performance Dashboard
- Fraud & Risk Monitoring Dashboard

Overall, this dataset provides a strong foundation for analyzing the credit card ecosystem, identifying spending patterns, assessing fraud risk, and supporting data-driven decision-making through visually rich and interactive Power BI reports.

3. Pre-Processing the Dataset

Before building dashboards and generating insights, the dataset underwent multiple preprocessing steps to ensure data quality, consistency, and accurate analysis. These steps were mainly carried out using **Power Query Editor** and **DAX** in Power BI.

1. Data Import and Integration

- Imported multiple CSV files from the Kaggle *Transactions & Fraud Dataset* into Power BI.
- Loaded fact and dimension tables such as:
 - transactions_flat
 - cards_dim
 - users_dim
 - merchantdim_trans
 - locationdim_user
 - personaldetailsdim_user
- Ensured proper data type detection during import.

2. Data Type Correction

- Converted columns to appropriate data types:
 - Dates → Date (e.g., transaction date, account open date)
 - Monetary fields → Decimal Number (Balance, Credit Limit, Spend)
 - IDs → Whole Number / Text
- This ensured accurate aggregations and time-based analysis.

3. Handling Missing and Null Values

- Identified missing or null values across numeric and categorical fields.

- Applied suitable treatments:
 - Removed rows with critical missing values (e.g., transaction amount).
 - Replaced nulls with default values where applicable.
 - Ensured no blank values affected KPIs and visuals.

4. Removing Duplicates

- Checked for duplicate records using:
 - transaction_id
 - card_id
 - client_id
- Removed duplicate entries to avoid inflated transaction counts and balances.

5. Feature Engineering (Derived Columns)

Several new columns were created to enhance analysis:

- **Age Group**
 - Categorized users into groups such as:
 - 20's, 30's, 40's, 50's, 50+
- **Income Category**
 - Grouped users based on per-capita income:
 - 50K, 100K, 150K, 200K, 250K, 300K, 350K
- **High Risk Flag**
 - Created a calculated column to classify users/cards as:
 - High Risk
 - Normal
 - Based on credit score and transaction behaviour.
- **Quarter**

- Extracted quarter information from transaction dates for trend analysis.

6. Categorical Data Standardization

- Standardized categorical values for consistency:
 - Gender → Male / Female
 - Card Type → Credit / Debit / Debit (Prepaid)
 - Card Brand → Visa, Mastercard, Amex, Discover
- Removed spelling inconsistencies and formatting issues.

7. Data Normalization and Aggregation

- Aggregated transaction-level data to:
 - Merchant-level revenue
 - State-wise revenue
 - Card-type-wise balance
- Ensured measures such as **Sum**, **Average**, and **Count** were correctly applied.

8. Relationship Modelling (Snowflake Schema)

- Established relationships between fact and dimension tables:
 - One-to-Many relationships for efficient filtering.
- Implemented a **snowflake schema** for:
 - Better performance
 - Accurate cross-filtering
 - Clean report structure
 - Data Normalization(Since the dataset is to big so normalization is very important to reduce the ambiguity in data).
 -

9. Measure Creation Using DAX

Created custom measures to support dashboard KPIs:

- Total Transactions
- Total Cards
- Average Credit Score
- Average Spend per Merchant
- Sum of Balance by Card Type
- Average Debit by Income Category

10. Data Validation and Quality Checks

- Cross-verified totals and aggregations across visuals.
- Ensured slicers (Age Group, Gender, Card Type, Income Category) worked correctly.
- Validated that filters applied consistently across all dashboard pages.

11. Optimization for Visualization

- Reduced unnecessary columns to improve performance.
- Renamed columns for better readability.
- Sorted categorical fields logically (Age Groups, Income Levels).

4. Model Building

In this Power BI project, a Snowflake Schema data model was designed to efficiently analyze large-scale credit card transaction data sourced from Kaggle. The snowflake schema was chosen to ensure data normalization, reduced redundancy, improved performance, and scalable analytics across customer behaviour, merchant performance, card usage, and fraud risk monitoring.

The data model consists of one central fact table connected to multiple dimension tables, where some dimensions are further normalized into sub-dimensions, forming a snowflake structure.

Central Fact Table:

1. transactions_flat (Fact Table)

This table acts as the core transactional fact table, storing detailed transaction-level information.

Key attributes:

- Trans_id – Unique transaction identifier
- Trans_date – Transaction date
- Quarter – Time-based aggregation
- Balance_\$ – Transaction amount (measure)
- card_id – Foreign key to card details
- client_id – Foreign key to customer details
- merchant_id – Foreign key to merchant details

Purpose:

- Stores all measurable values used in KPIs
- Enables analysis of transaction volume, balances, trends, and fraud indicators

Dimension Tables:

2. cards_dim (Card Dimension)

Contains information related to payment cards.

Attributes:

- card_id (Primary Key)
- card_brand (Visa, Mastercard, Amex, Discover)

- card_type (Credit, Debit, Debit Prepaid)
- credit_limit_\$
- has_chip
- acct_open_date
- expires

Relationship:

- One-to-Many with transactions_flat via card_id

Business Insights Enabled:

- Card performance analysis
- Credit limit vs spending behavior
- Chip-enabled vs non-chip transactions
- Risk analysis by card type and brand

3. users_dim (Customer Dimension)

Stores financial and demographic details of customers.

Attributes:

- client_id (Primary Key)
- credit_score
- current_age
- gender
- Income Cat
- num_credit_cards
- per_capita_income_\$
- High Risk Flag

Relationship:

- One-to-Many with transactions_flat via client_id

Business Insights Enabled:

- Customer risk profiling
- Income vs spending analysis
- Credit score-based segmentation
- Fraud risk identification

4. PersonalDetailsDim_user (Sub-Dimension – Snowflake Extension)

This table further normalizes customer personal data.

Attributes:

- id (Primary Key)
- birth_year
- birth_month
- current_age
- gender
- Age Group

Relationship:

- One-to-One with users_dim

Purpose:

- Supports age-group-based analytics
- Reduces redundancy in the main user table
- Enables demographic-level insights

5. LocationDim_user (Sub-Dimension – Snowflake Extension)

Stores geographical details of customers.

Attributes:

- id (Primary Key)
- address

- latitude
- longitude

Relationship:

- One-to-One with users_dim

Business Insights Enabled:

- Location-based transaction mapping
- Geographic fraud pattern analysis
- Regional customer behavior trends

6. Merchantdim_trans (Merchant Dimension)

Contains merchant-related details.

Attributes:

- merchant_id (Primary Key)
- merchant_city
- merchant_state
- zip
- mcc (Merchant Category Code)

Relationship:

- One-to-Many with transactions_flat via merchant_id

Business Insights Enabled:

- Top merchants by revenue
- State-wise and city-wise merchant analysis
- Spend category insights using MCC

Why Snowflake Schema Was Used

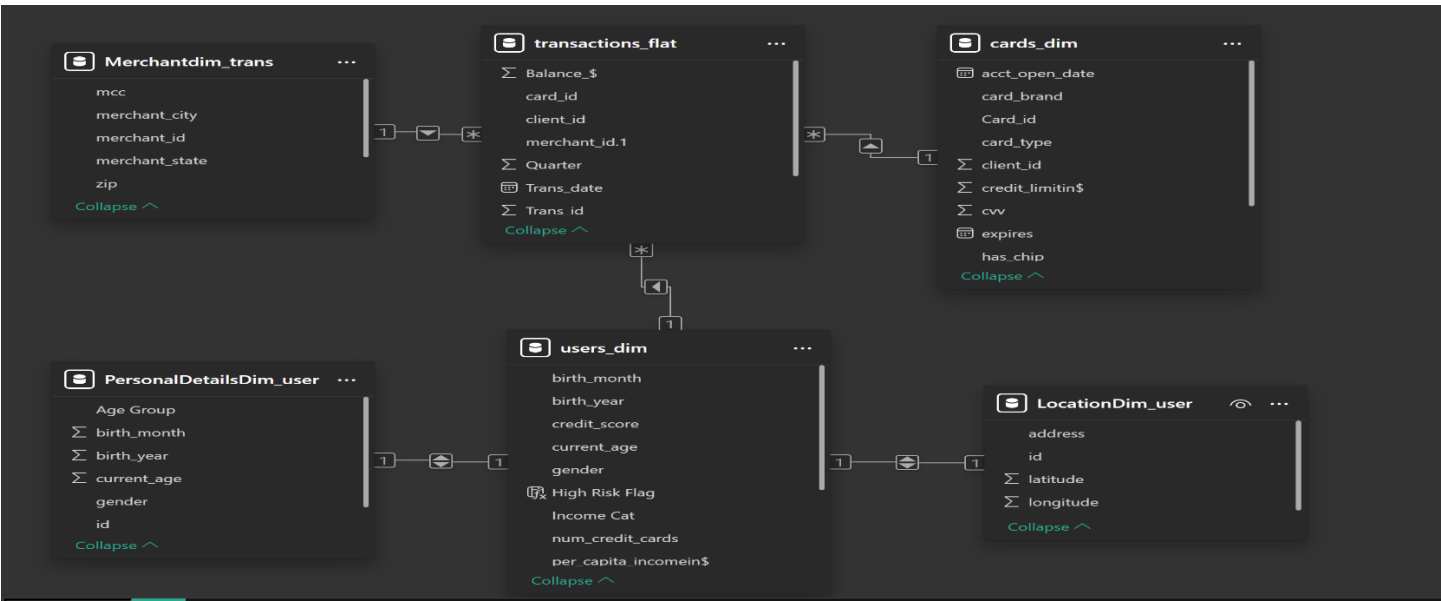
The Snowflake Schema was intentionally selected due to the following advantages:

- **Reduced Data Redundancy:** Personal and location details are separated into sub-dimensions
- **Improved Data Integrity:** Normalized tables maintain consistent attribute values
- **Scalability:** Easy to add new dimensions without restructuring the model
- **Optimized Performance:** Fact table remains lean with only keys and measures
- **Better Analytics:** Enables deep drill-downs across customer, merchant, card, and geography.

Screenshot of Relations:

Relationship Summary		
From Table	To Table	Relationship
transactions_flat	cards_dim	Many-to-One
transactions_flat	users_dim	Many-to-One
transactions_flat	merchantdim_trans	Many-to-One
users_dim	PersonalDetailsDim_user	One-to-One
users_dim	LocationDim_user	One-to-One

Screenshot of Data-model:



5. Analysis on dataset

Introduction

The analysis phase focuses on extracting meaningful business insights from the credit card transaction dataset using Power BI. This stage bridges data preprocessing and dashboard development by examining customer behaviour, transaction patterns, card performance, merchant activity, and fraud risk indicators. The analysis combines statistical aggregation and visual exploration to understand data trends, identify anomalies, and support data-driven decision-making for banking and financial services use cases.

i. General Description

The dataset consists of large-scale credit card transaction data integrated with customer, card, merchant, and location dimensions. It includes both numerical and categorical variables that collectively represent the end-to-end credit card ecosystem.

Key attributes analyzed include:

1. Transaction Metrics

- Balance_\$ (transaction amount)
- Transaction count
- Transaction date and quarter

2. Customer Attributes

- Age and Age Group
- Gender
- Income Category
- Credit Score
- Number of Credit Cards

3. Card Attributes

- Card Brand (Visa, Mastercard, Amex, Discover)
- Card Type (Credit, Debit, Debit Prepaid)
- Credit Limit
- Chip Availability (has_chip)

4. Merchant Attributes

- Merchant ID
- Merchant State and City
- Merchant Category Code (MCC)

5. Risk Indicators

- High Risk Flag
- Credit score variations
- Transaction behaviour patterns

Outcome Variable and Analytical Focus

Instead of a single predictive target variable, this project focuses on descriptive and diagnostic analytics using key outcome indicators such as:

- Total and average transaction balance
- High Risk Flag (Normal vs High Risk customers/cards)
- Merchant revenue contribution
- Customer segmentation by age, income, and credit score
- Card performance by type and brand

The High Risk Flag serves as a critical analytical indicator to classify customers and cards into Normal and High Risk categories, supporting fraud and risk monitoring use cases.

Key Analytical Insights Explored

- **Customer Behaviour Analysis**
 - Spending patterns across age groups, gender, and income categories
 - Relationship between income level, credit score, and transaction value
- **Card Performance Analysis**
 - Balance and credit limit comparison across card types and brands
 - Usage trends of credit, debit, and prepaid cards
 - Impact of chip-enabled cards on transaction volume

- **Merchant and Geographic Analysis**
 - Top states and merchants contributing the highest revenue
 - Location-based transaction concentration using map visuals
 - Merchant category-wise revenue distribution
- **Fraud and Risk Analysis**
 - Comparison of credit scores between normal and high-risk customers
 - Identification of card brands and card types with higher risk exposure
 - Transaction patterns associated with high-risk flags

Analytical Approach

The dataset was analyzed using:

- **Aggregations** (Sum, Average, Count) for financial metrics
- **Cross-dimensional slicing** using Age Group, Gender, Card Type, Income Category
- **Visual analytics** such as bar charts, pie charts, tables, and maps
- **DAX-based KPIs** to ensure consistent and accurate calculations

This analysis validated the quality and relevance of the dataset for business intelligence applications and laid the foundation for building interactive dashboards such as:

- Customer Insight Dashboard
- Merchant & Spend Category Analytics Dashboard
- Card Performance Dashboard
- Fraud & Risk Monitoring Dashboard

Overall, the analysis phase enabled a comprehensive understanding of the credit card ecosystem, helping uncover customer trends, revenue drivers, and potential fraud risks through structured Power BI analytics.

6. Analysis of Dashboards

The analysis was conducted through multiple Power BI dashboards, each focusing on a specific analytical theme:

1. Customer Insight Analysis

This analysis focuses on understanding customer demographics and spending behaviour.

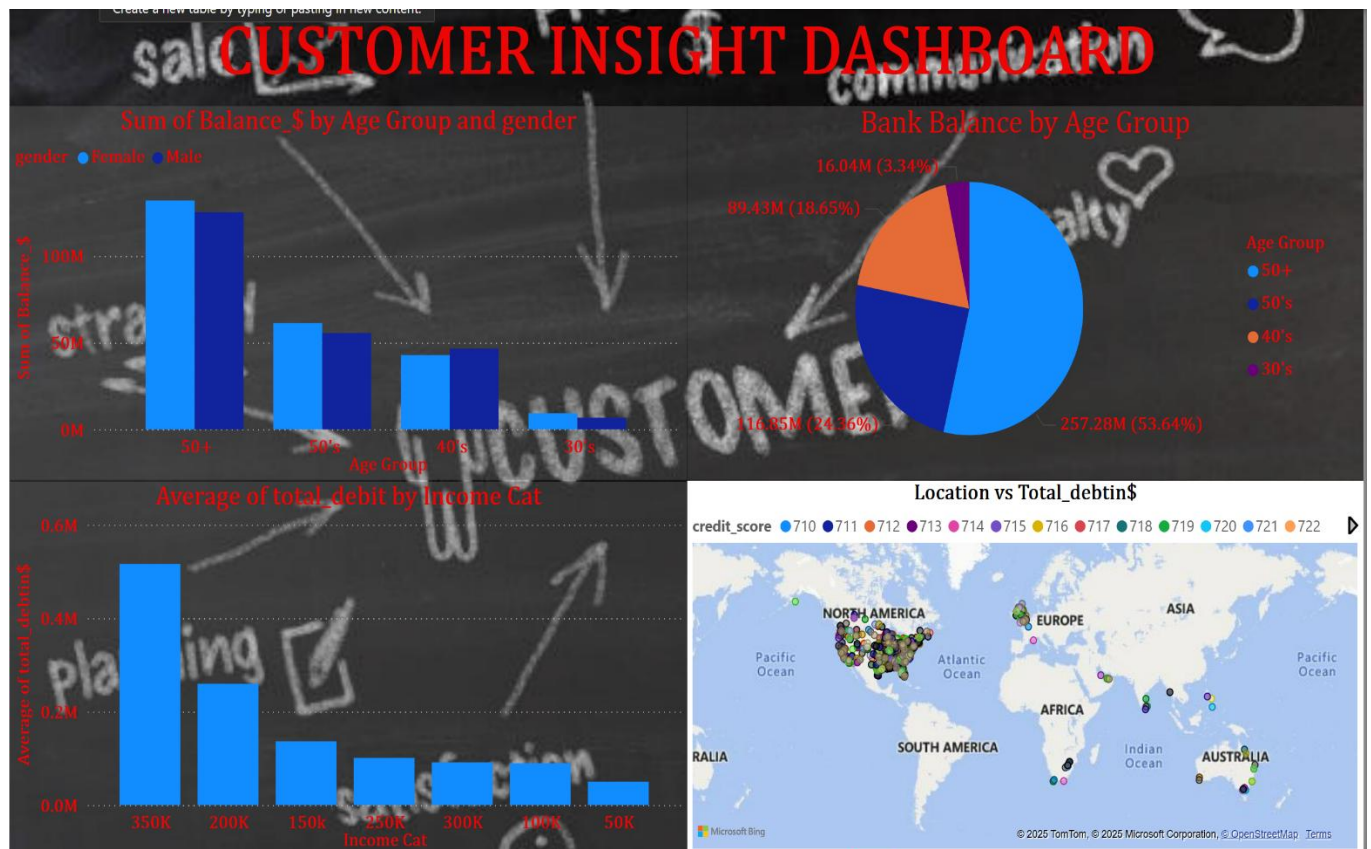
Key Insights Extracted:

- Analyzed sum of transaction balances by age group and gender to identify high-spending demographics.
- Evaluated average debit amount by income category, revealing that higher income groups tend to have higher average spending.
- Studied credit score distribution across customers to understand financial health patterns.
- Used geographical maps to visualize customer transaction concentration across regions.

Business Value:

- Helps banks identify high-value customer segments
- Supports targeted marketing and personalized financial products

Screenshot Of Dashboard:



2. Merchant & Spend Category Analysis

This dashboard analyzes merchant performance and revenue contribution.

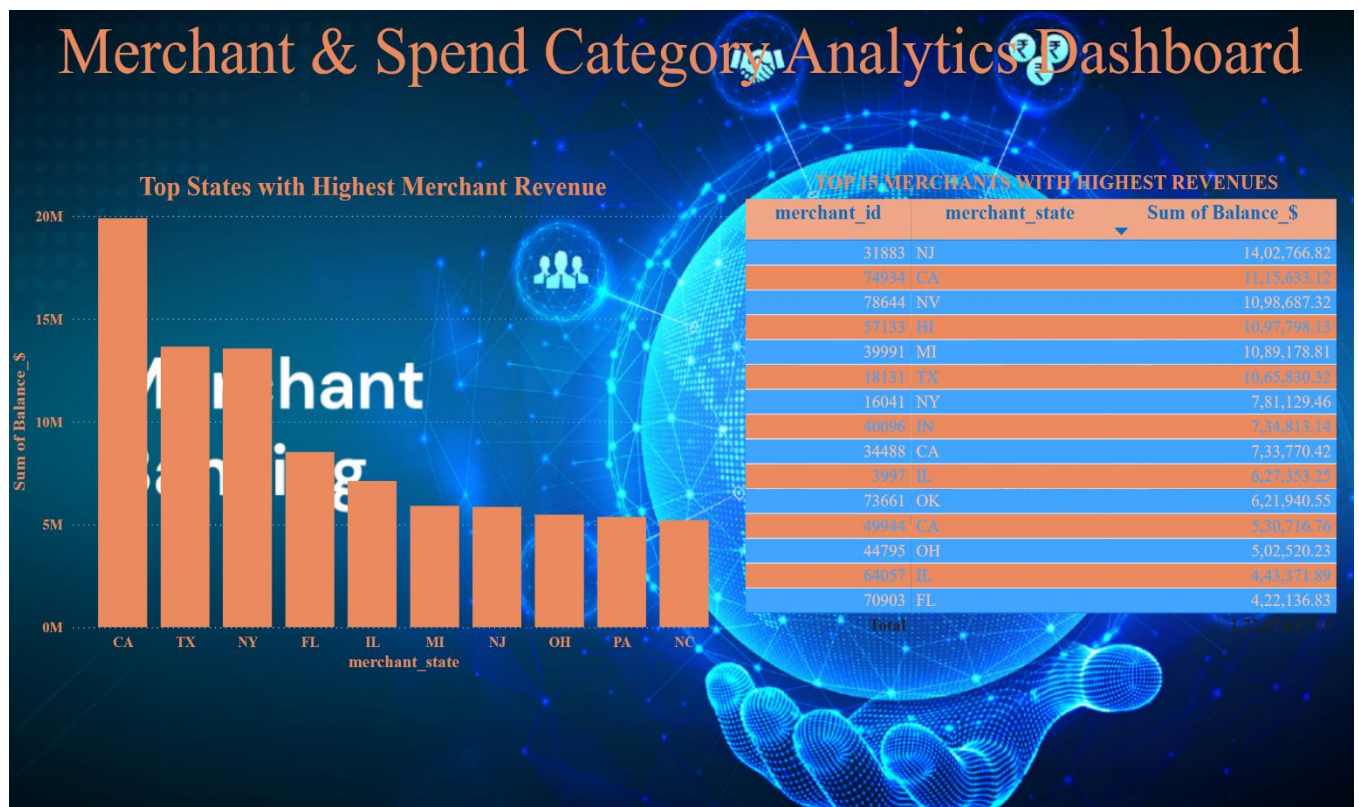
Key Insights Extracted:

- Identified top states with highest merchant revenue, highlighting regional business hotspots.
- Ranked top 15 merchants contributing maximum balance, enabling merchant performance benchmarking.
- Analyzed merchant category codes (MCC) to understand spending behaviour across business categories.

Business Value:

- Supports merchant partnership strategies
- Identifies high-revenue locations and industries

Screenshot Of Dashboard:



3. Card Performance Analysis

This analysis evaluates how different card types and brands perform.

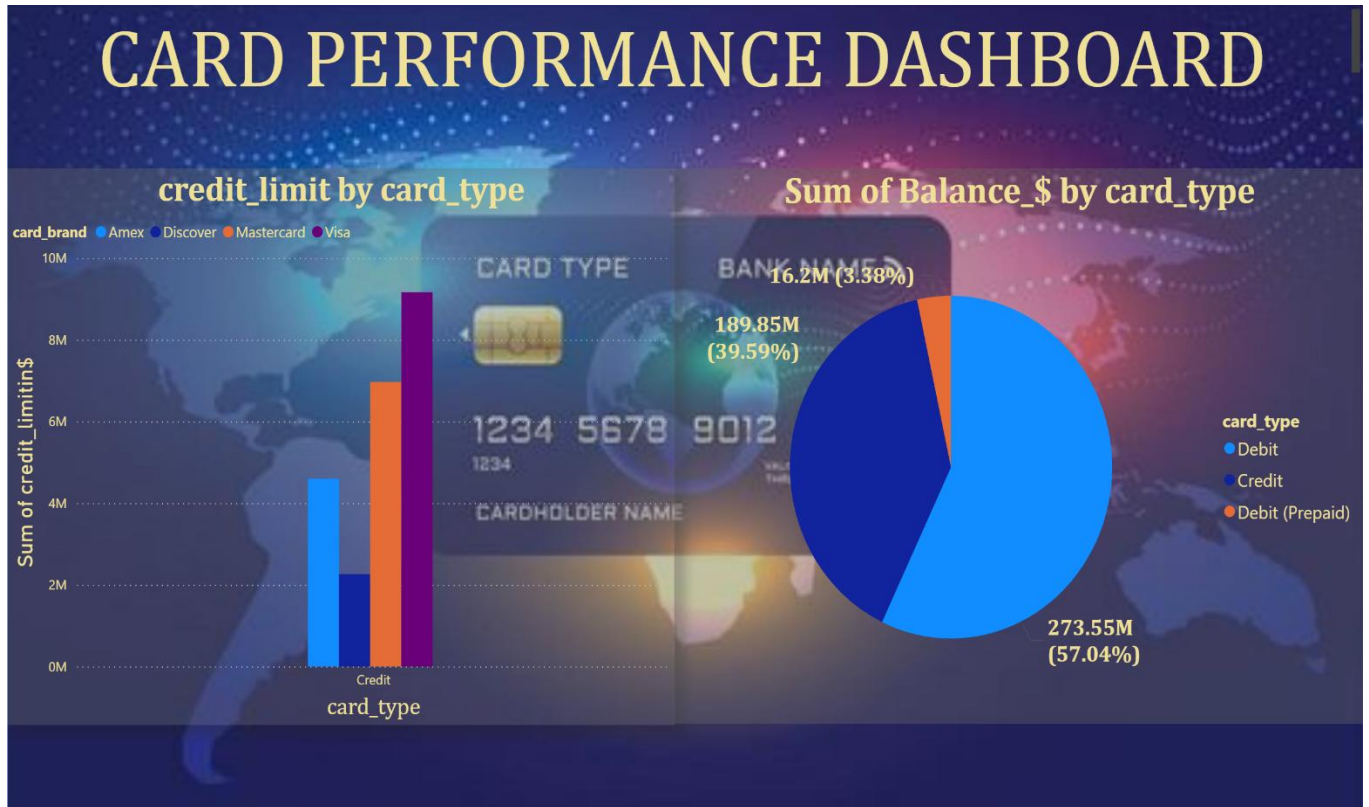
Key Insights Extracted:

- Compared credit limits by card type and card brand.
- Analyzed balance distribution across credit, debit, and prepaid cards.
- Assessed performance differences among Visa, Mastercard, Amex, and Discover.

Business Value:

- Helps optimize card product offerings
- Identifies customer preference trends across card types

Screenshot Of Dashboard:



4. Fraud & Risk Monitoring Analysis

This dashboard focuses on identifying potential fraud and high-risk patterns.

Key Insights Extracted:

- Compared transactions made using chip-enabled vs non-chip cards.
- Analyzed average credit score by high-risk flag, highlighting risk-prone customer segments.
- Identified card brands and card types frequently flagged as high risk.
- Studied relationships between credit score, card usage, and fraud indicators.

Business Value:

- Enables proactive fraud monitoring
- Supports risk-based decision making and credit control

Screenshot Of Dashboard:



iii. Analytical Outcome

The analysis confirms that the dataset is highly suitable for business intelligence and risk analytics. The dashboards effectively reveal:

- Clear spending patterns across demographics and income groups
- Significant variation in merchant revenue by state and category
- Distinct performance differences across card types and brands
- Strong correlation between credit score, transaction behaviour, and fraud risk

By leveraging a snowflake schema data model, Power BI ensures accurate aggregations, efficient filtering, and seamless drill-down analysis across all dashboards.

Overall, the analysis phase successfully transforms raw transactional data into meaningful insights that support revenue optimization, customer segmentation, fraud prevention, and strategic decision-making in the credit card ecosystem.

5. Credit Card Ecosystem Analytics Dashboard

The Credit Card Ecosystem Analytics Dashboard provides a consolidated and high-level view of customer behaviour, card usage patterns, merchant performance, and financial risk indicators within the credit card ecosystem. This dashboard integrates insights from multiple dimensions customers, cards, merchants, income categories, demographics, and transactions using interactive visuals and KPIs to support data-driven decision-making.

The dashboard is designed to enable stakeholders to quickly monitor key performance metrics, identify spending trends, assess risk exposure, and analyze revenue contribution across different customer and

merchant segments.

Key Performance Indicators (KPIs)

At the top of the dashboard, key KPIs provide a snapshot of the overall system performance:

- **Total Cards (6146)**
Represents the total number of active cards issued in the dataset.
- **Average Credit Score (709.73)**
Indicates the overall creditworthiness of customers, useful for risk profiling.
- **Total Transactions (12M)**
Shows the total transaction volume processed, highlighting system scale and usage intensity.
- **Average Spend per Merchant (41.16)**
Reflects the average transaction value across merchants, supporting merchant performance evaluation.

Customer Spending & Demographic Analysis:

Average Total Debit by Income Category

This bar chart analyzes average debit spending across income categories (50K to 350K):

- Higher income groups generally show higher average debit usage.
- Helps understand spending capacity and customer value segmentation.
- Useful for targeted marketing and personalized financial offerings.

Sum of Balance by Age Group and Gender

This clustered bar chart compares total balance across age groups (30s, 40s, 50s, 50+) and gender:

- Customers aged 50+ contribute the highest balance, indicating strong purchasing power.
- Gender-based comparison highlights spending differences between male and female customers.
- Supports demographic-based financial product strategies.

Card Usage Analysis:

Sum of Balance by Card Type

This pie chart represents balance distribution across:

- Debit
- Credit
- Debit (Prepaid)

Key insights:

- Debit cards contribute the largest share of total balance.
- Credit cards show significant usage, reflecting borrowing and repayment behavior.
- Prepaid cards account for a smaller but relevant portion.

This analysis helps assess card performance and usage preferences.

Merchant Performance Analysis:

Top States with Highest Merchant Revenue

This bar chart highlights states generating the highest merchant revenue:

- States like CA, TX, and NY dominate transaction balances.
- Enables geographic revenue concentration analysis.
- Supports regional business expansion and merchant targeting decisions.

Top 15 Merchants with Highest Revenue

The table lists the top 15 merchants based on total transaction balance:

- Displays merchant ID, merchant state, and revenue contribution.
- Helps identify high-value merchants and strategic partnerships.
- Useful for merchant-focused risk and performance monitoring.

Risk & Fraud Monitoring:

Cards and Their Risk Flags

This table visual displays card brand, card type, and high-risk flag status:

- Identifies card combinations frequently associated with high-risk activity.
- Supports proactive fraud monitoring.
- Helps financial institutions prioritize risk mitigation strategies by card type and brand.

Interactive Filtering & Slicers

The dashboard includes slicers for:

- Age Group
- Income Category
- Card Type
- Gender

These filters allow users to:

- Drill down into specific customer segments
- Perform comparative analysis
- Dynamically explore trends across demographics and financial attributes

Business Value of the Dashboard

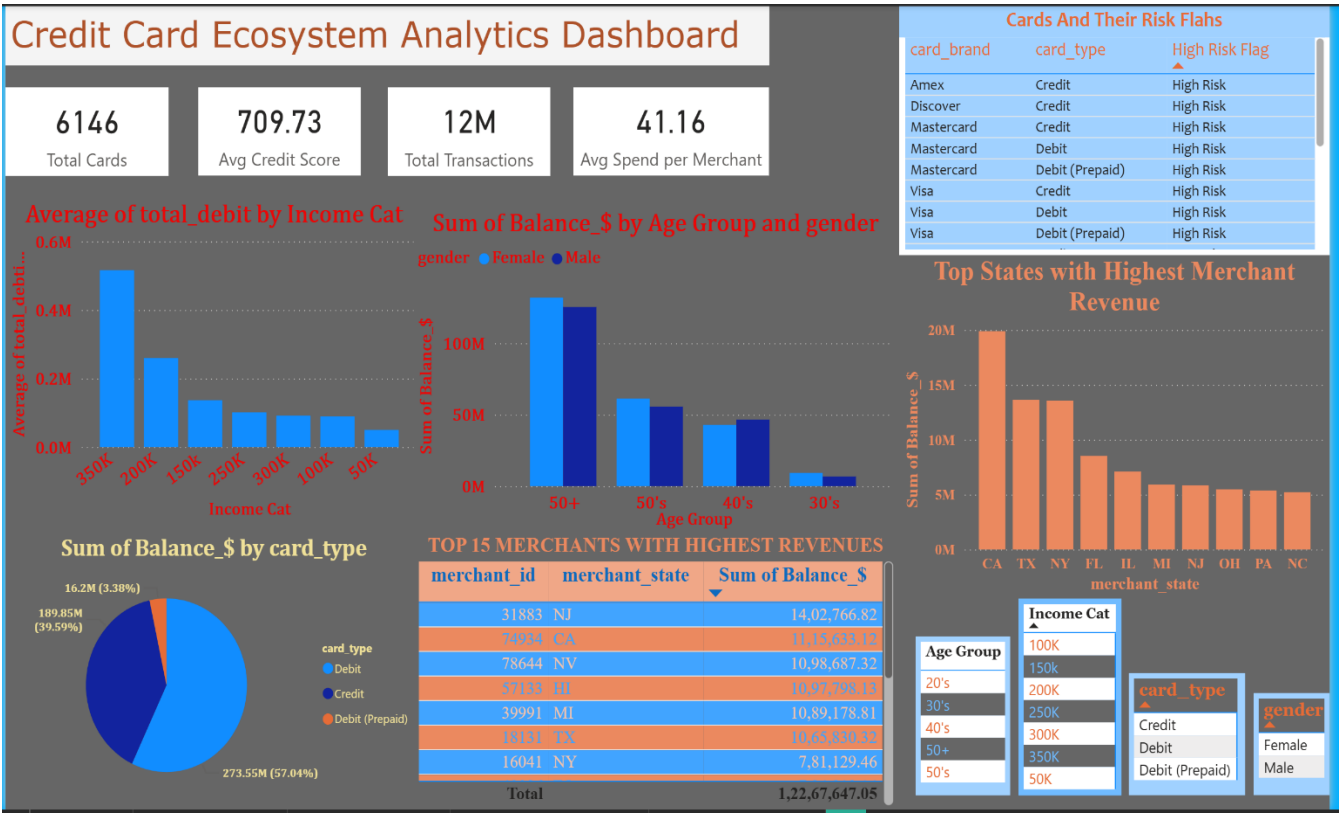
- Provides a 360-degree view of the credit card ecosystem
- Enables customer segmentation, revenue optimization, and risk monitoring

- Supports data-driven strategic decisions in banking and financial services
- Demonstrates effective use of Power BI, DAX measures, and a snowflake schema data model

Conclusion

The Credit Card Ecosystem Analytics Dashboard successfully combines customer insights, card performance metrics, merchant revenue analysis, and risk indicators into a single interactive Power BI solution. It allows stakeholders to understand spending behaviour, identify high-value customers and merchants, and proactively monitor financial risk, making it a powerful tool for business intelligence and operational decision-making.

Screenshot Of Dashboard:



7. Conclusion

This project successfully demonstrates the application of business intelligence and data analytics using Power BI to analyze a large-scale credit card transaction dataset. By integrating transaction data with customer demographics, card attributes, merchant information, and geographic details, the project delivers a comprehensive view of the credit card ecosystem.

The use of a snowflake schema data model ensured efficient data organization, reduced redundancy, and accurate cross-table analysis. Interactive dashboards provided deep insights into customer spending behaviour, card performance, merchant revenue trends, and fraud risk patterns. Key findings highlighted how factors such as age group, income category, card type, and credit score significantly influence spending and risk profiles.

Overall, the project showcases how data-driven dashboards can support strategic decision-making in the banking and financial services domain by enabling proactive risk monitoring, customer segmentation, and revenue optimization.

8. Future Scope

The project can be further enhanced in the following ways:

1. Integration of Machine Learning Models
 - Incorporate predictive models for fraud detection and customer risk scoring.
2. Real-Time Data Integration
 - Connect Power BI with live transaction streams for real-time fraud monitoring.
3. Advanced Time-Series Analysis
 - Perform trend forecasting on transaction volume and merchant revenue.
4. Behavioural Segmentation
 - Apply clustering techniques to identify high-value and high-risk customer groups.
5. Enhanced Fraud Indicators
 - Include additional fraud signals such as transaction velocity, unusual location patterns, and device-level data.
6. Enterprise Deployment
 - Publish dashboards to Power BI Service with role-level security for enterprise use.
7. Automated Alerts
 - Set up alerts for abnormal transaction patterns or high-risk threshold breaches.

9. Reference

1. Kaggle Dataset – *Transactions Fraud Datasets*
<https://www.kaggle.com/datasets/computingvictor/transactions-fraud-datasets>
2. Microsoft Power BI Documentation
<https://learn.microsoft.com/en-us/power-bi/>
3. SQL & Data Modelling Concepts
<https://learn.microsoft.com/en-us/sql/relational-databases/>
4. Data Warehousing Concepts – Star & Snowflake Schema
Kimball, R. & Ross, M., *The Data Warehouse Toolkit*
5. Credit Risk & Fraud Analytics Concepts
Hand, D. J., & Whitrow, C., *Fraud Detection in Financial Services*